UTKALSARATHI: AI-POWERED SMART TOURISM, CULTURAL DISCOVERY AND TRAVELLER SAFETY PLATFORM FOR ODISHA

1. PROJECT OVERVIEW

UtkalSarathi is a proposed AI-powered, multilingual smart tourism platform designed to make exploring Odisha easier, more accessible, culturally enriching, and safer through informed travel decisions. It brings destination discovery, personalized itinerary planning, local transportation, transparent travel-cost estimates, accommodation, regional cuisine, cultural events, historical storytelling, traveller assistance, and local business discovery into one connected digital ecosystem.

The platform addresses the challenges faced by tourists who rely on fragmented or unreliable information, particularly when travelling to lesser-known destinations such as eco-tourism camps, tribal heritage regions, traditional villages, and nature reserves. By combining conversational AI with curated tourism information, mapping services, community-based reporting, and offline access to selected resources, UtkalSarathi aims to help travellers plan complete journeys and discover experiences beyond Odisha's most prominent destinations.

2. PROBLEM UNDERSTANDING AND PROPOSED SOLUTION

Travellers visiting Odisha often need multiple sources to discover attractions, identify suitable transport, estimate local travel costs, find accommodation, explore regional food, and learn about cultural events. These difficulties become more pronounced at remote or lesser-known locations, where last-mile transport information, regional-language guidance, local services, and reliable connectivity may be limited.

This information gap can result in inefficient routes, unexpected expenses, language barriers, missed cultural experiences, and difficulties making informed decisions in unfamiliar places. Generic navigation tools may not provide the local cultural context and integrated travel planning needed for a complete visitor experience.

UtkalSarathi proposes a mobile-first, AI-powered travel companion that connects these services. Travellers will be able to enter their interests, budget, available time, starting location, and preferred language to receive personalized itineraries, transport options, estimated costs, cultural recommendations, and relevant local services. A WhatsApp-based conversational interface may also be explored, subject to integration feasibility, platform requirements, and available development resources. The platform will distinguish verified details from estimates and unverified reports, and provide offline access to selected resources.

3. CORE FEATURES AND INNOVATION

A. AI-Powered Personalized Travel Planning

The platform will generate itineraries based on travellers' interests, budget, available time, starting location, and preferred experiences. Recommendations will cover prominent attractions as well as lesser-known heritage sites, eco-tourism locations, nature reserves, traditional villages, and community-based cultural experiences. The planner will consider journey duration, available destination information, transport options, verified opening hours where available, and estimated expenses. It will help users compare alternatives and adjust plans when their time or budget changes.

B. Intelligent Local Transportation and Transparent Pricing

UtkalSarathi will help travellers explore available government buses, participating private transport services, taxis, and auto-rickshaws, depending on accessible information. Mapping and routing integrations will provide distances, estimated journey times, route options, and available transport details. Fares and timetables will show their source or last-updated status where available. Estimated prices will be distinguished from verified or provider-confirmed prices. Special attention will be given to last-mile connectivity from major railway stations, bus stops, and other transport hubs to smaller attractions. Where reliable information is unavailable, the platform will communicate that limitation rather than inventing routes, timings, or fares. Live tracking and real-time updates may be introduced through authorized data feeds and provider partnerships.

C. Multilingual Conversational AI Travel Assistant

The assistant will answer questions about destinations, customs, transport, accommodation, cuisine, cultural events, and travel budgets in supported languages, initially prioritizing Odia, Hindi, and English. It will use curated tourism information and trusted sources to provide contextual answers and explain recommendations. It will acknowledge uncertainty when information cannot be verified. Voice input and spoken responses may improve accessibility, with additional languages and interaction capabilities introduced progressively.

D. UtkalSuraksha: Traveller Safety and Assistance

UtkalSuraksha will provide verified emergency contact information, an SOS calling shortcut, optional location sharing with trusted contacts, and discovery of nearby hospitals, police stations, and relevant assistance services. Where reliable and sufficiently current official sources are available, the platform will display weather warnings, severe-weather advisories, road closures, and travel disruptions with source and update information. Users may submit reports of local hazards and travel disruptions. Official advisories will be distinguished from community reports, and moderation, timestamps, verification status, and expiry mechanisms will help reduce misinformation and outdated alerts. The system will clearly communicate network, GPS, and permission requirements. An SOS calling shortcut will not be represented as a confirmed alert to authorities unless a real integration and delivery confirmation exist. Safety guidance will not replace official emergency services.

E. Cultural Discovery and AI-Powered Historical Voice Stories

The platform will showcase Odisha's temples, historical monuments, tribal traditions, handloom, handicrafts, regional cuisine, festivals, traditional villages, and local markets. AI-assisted audio guides will explain the history, architecture, cultural significance, and documented stories of selected destinations such as Konark Sun Temple, Jagannath Temple, Udayagiri and Khandagiri Caves, and Dhauli Shanti Stupa. Travellers will be able to listen in supported languages and ask follow-up questions about monuments, historical personalities, ancient kingdoms, architecture, and traditions. Documented information will be distinguished from traditional stories and legends, using curated sources and appropriate review. Cultural events such as Ratha Yatra and Nuakhai will be listed with dates and source details where reliable information is accessible. Selected guides will be downloadable for offline use, and QR-based access may support destination-specific stories.

F. Accommodation, Regional Cuisine and Local Business Integration

The platform will connect travellers with participating hotels, resorts, eco-tourism camps, restaurants, guides, artisans, handloom producers, and handicraft businesses. Users will be able to explore listed services, compare available prices, discover regional cuisine, and access enquiry or booking options through participating providers and supported services. A business interface may allow participating providers to update descriptions, contact information, menus, prices, and availability. Verification labels

and reporting mechanisms will help improve information quality and visibility for smaller businesses and community-led tourism experiences.

G. Budget-Friendly Travel and Offline Access

Travellers will be able to plan around their budget, available time, group size, and preferred destinations. Estimates may include transport, food, accommodation, and entry fees where information exists. Important assumptions and excluded costs will be identified. Offline mode will support saved itineraries, selected destination details, cultural stories, and essential emergency contact information. Live data, online bookings, location sharing, and other connection-dependent features will clearly state their requirements.

4. TECHNICAL APPROACH AND CURRENT DEVELOPMENT STAGE

UtkalSarathi will combine AI services with conventional software components. The proposed architecture includes a mobile-friendly interface, backend services, a structured tourism database, mapping and routing integrations, a conversational AI assistant, and a retrieval-based knowledge system for source-grounded tourism and historical answers. AI will support natural-language interaction, recommendations, and cultural explanations. Routing tools and deterministic calculations will handle journey estimates, itinerary constraints, and budget arithmetic.

The platform will use curated destination records, available official tourism information, participating business listings, and authorized transport, weather, and event data where accessible. Appropriate source, freshness, and verification indicators will help users assess information reliability.

CURRENT DEVELOPMENT STAGE UtkalSarathi is at the concept and proposal stage; a functional prototype has not yet been developed. The initial implementation will focus on a curated selection of destinations and a connected traveller journey: entering preferences, generating an itinerary, reviewing transport options and estimated costs, exploring a cultural story, discovering local services, and accessing saved travel and safety information.

5. PHASED IMPLEMENTATION ROADMAP

Phase 0 — Define and prepare: Confirm the MVP scope and user journey; identify data sources and owners; curate initial records for approximately 10–15 destinations; prepare low-fidelity screens, historical source lists, and test cases. Gate: the scope and seed data are reviewed before development expands.

Phase 1 — Initial MVP: Build itinerary planning with time and budget constraints; route options and labelled cost estimates; a source-grounded tourism Q&A assistant; 3–5 reviewed cultural story guides; an initial Odia, Hindi, and English content path; and saved guides plus essential emergency contacts. Gate: one end-to-end traveller journey works with the curated data, and the team can explain its limitations.

Phase 2 — Trusted integrations: Add official weather and event information where reliable feeds are accessible, a moderated hazard-report workflow, local business listing management, downloadable offline packs, and a feasibility test for WhatsApp access if worthwhile. Gate: source freshness, permissions, and report-handling workflows are tested.

Phase 3 — Scale after validation: Explore authorized live transport feeds, booking-provider integrations, broader destination and language coverage, tourism and community partnerships, and usage analytics. Gate: expansion depends on pilot feedback, data availability, partner readiness, and demonstrated reliability.

Features requiring external partnerships, live data feeds, automated emergency alerts, or extensive statewide coverage will be introduced progressively as the necessary data, permissions, and integrations become available. Proposed phases are deliverables, not claims that work has already been completed.

6. INNOVATION AND DIFFERENTIATION

UtkalSarathi's key innovation is connecting personalized itinerary planning, last-mile transport information, transparent cost estimates, multilingual cultural storytelling, safety-aware guidance, offline resources, and local business discovery in one tourism experience. Instead of functioning only as a destination directory or generic navigation tool, it aims to help travellers understand how to complete a journey, what it may cost, which local experiences are available, how reliable the information is, and which alternatives they can consider. Its emphasis on lesser-known destinations, source-grounded AI, practical connectivity limitations, and local participation aims to extend digital tourism beyond major tourist centres.

7. SOCIAL IMPACT AND SUSTAINABLE TOURISM

UtkalSarathi aims to improve access to tourism information, encourage responsible exploration, and strengthen connections between visitors and local communities. Expected benefits include easier trip planning, discovery of lesser-known attractions, better access to cultural information, more transparent travel costs, and greater visibility for local transport operators, accommodation providers, food businesses, artisans, guides, and rural tourism initiatives. The platform will promote culturally respectful and environmentally responsible tourism, especially when presenting community-sensitive heritage locations and ecologically fragile destinations. Impact can be assessed through itinerary completion, user satisfaction, quality of cost estimates, use of cultural guides, discovery of lesser-known destinations, and referrals to participating local businesses.

8. BUSINESS MODEL AND FUTURE SCALABILITY

The proposed business model includes booking commissions through participating providers, premium business listings, clearly labelled promotional services, and partnerships with tourism operators and local enterprises. Commercial recommendations will be distinguished from independent travel guidance, and essential safety information will remain separate from paid promotion. Future development may include verified live bus tracking, expanded transport and booking integrations, more regional languages, enhanced offline assistance, real-time cultural event updates, accessibility improvements, feasible WhatsApp access, and wider partnerships with tourism authorities, transport operators, and community organizations. The platform will begin with a manageable set of destinations and validated sources, then expand as user feedback, partnerships, and data availability support broader coverage.

9. VISION

Our vision is to make Odisha easier to explore through a trusted, intelligent, multilingual, and culturally responsible digital travel companion. By connecting tourism information, local transportation, cultural heritage, traveller assistance, and local enterprises, UtkalSarathi aims to create a more seamless visitor experience while supporting inclusive tourism growth and wider economic opportunities for Odisha's communities.